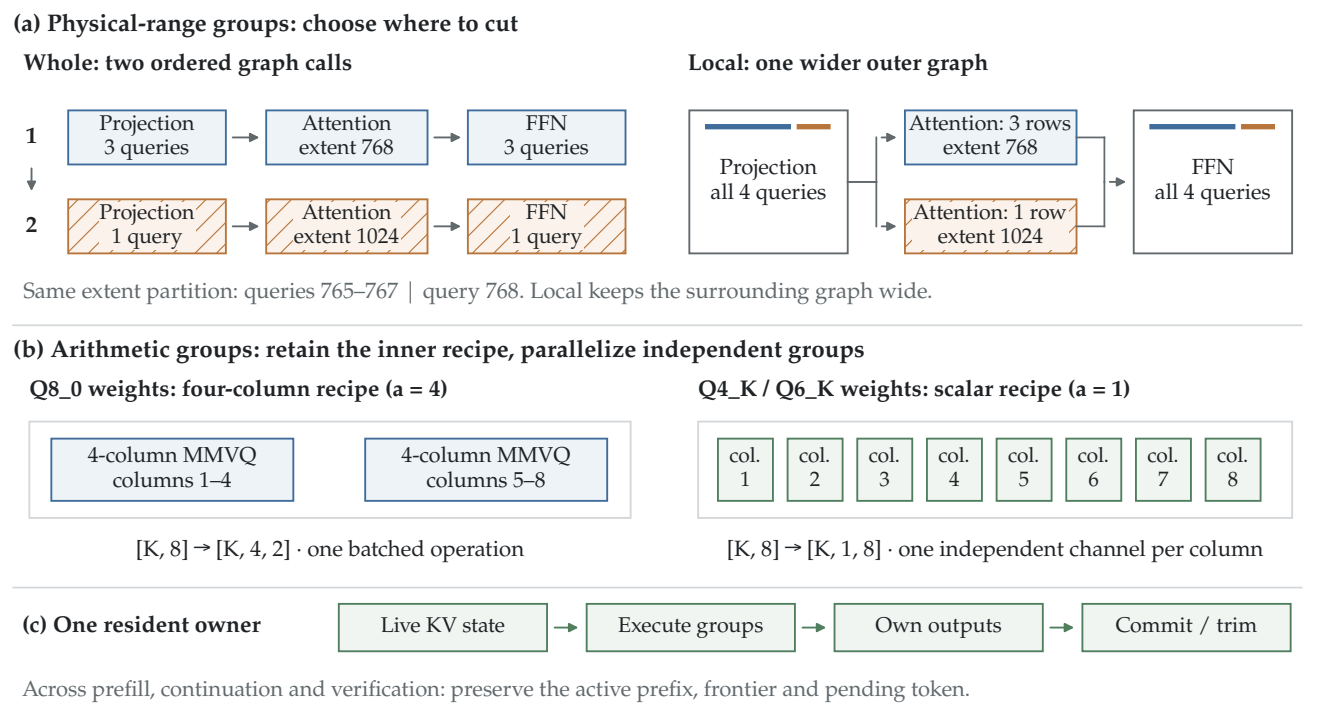

(a) Physical-range groups: choose where to cut
Whole: two ordered graph calls
1
2
Projection 3 queries
Attention extent 768
FFN 3 queries
Projection 1 query
Attention extent 1024
FFN 1 query
Local: one wider outer graph
Projection all 4 queries
Attention: 3 rows extent 768
Attention: 1 row extent 1024
FFN all 4 queries
Same extent partition: queries 765–767 | query 768. Local keeps the surrounding graph wide.
(b) Arithmetic groups: retain the inner recipe, parallelize independent groups
Q8_0 weights: four-column recipe (a = 4)
4-column MMVQ columns 1–4
4-column MMVQ columns 5–8
[K, 8] → [K, 4, 2] · one batched operation
Q4_K / Q6_K weights: scalar recipe (a = 1)
col. 1
col. 2
col. 3
col. 4
col. 5
col. 6
col. 7
col. 8
[K, 8] → [K, 1, 8] · one independent channel per column
(c) One resident owner
Live KV state
Execute groups
Own outputs
Commit / trim
Across prefill, continuation and verification: preserve the active prefix, frontier and pending token.